Assignment 1: AI Problem Solving

Complete

1/27/2026

Attempt 1

Review Feedback
1/27/2026

Attempt 1 Score:
Complete

View feedback

Anonymous grading: **no**

Unlimited Attempts Allowed

Details

General instructions

Read the whole module before you begin and manage your time between the tasks. Make reasonable assumptions as appropriate. Short and clear answers are preferred. Both members of the group must actively participate in every subtask. Each of you have up to 20 hours for a module including lectures.

All the problems in this assignment should be solved and handed in as a group, but you should be prepared to answer questions individually about your solutions. The full set of solutions should be submitted as a single PDF document in Canvas. When you submit code, you should submit both a PDF document describing your solution *as well as* code.

AI problem solving

In this module we will consider not just machine learning techniques but also some other basic models and methods which are often used to create applications with "intelligent" behaviour. The objective is to highlight the importance of selecting the overall approach and better understand when different techniques are applicable. In the last task we will also provide an overview of AI applications in industry.

A characteristic of several of the problems in this module is that they can be sensibly solved in different ways. Some problems also highlight that there is sometimes a fuzzy line between data science and AI applications.

For each problem you are asked to consider two main questions. First try to characterize each problem and how it can be modelled, for example as a:

- Function (classification, regression, ...)
- Optimization problem (constrained/unconstrained, ...)
- Dynamic system
- Discrete algorithmic problem

- Constraint satisfaction problem (e.g. constraints for colouring a map)
- ...

Additionally note other relevant characteristics of the problem.

Then - as appropriate - consider possible ways to solve:

- Statistics and/or machine learning
- Prescriptive modelling (modelling an idea). *(Explanation: The most common type of modelling is to describe something in reality, for example to simulate or predict. However, sometimes you define a score, e.g. BMI (for weight) or GINI (for poverty). Creating this score is then not just about straightforwardly modelling something in our physical world, but to quantify something according to what you consider as important.)*
- Discrete algorithm design (divide & conquer, dynamic programming, greedy, local search, other forms of search)
- Simulation
- Optimization, e.g. network flow, linear programming etc.
- Constraint programming (see reading task below)
- ...

While it is hardly possible to solve all these problems in detail given the limited time, focus on suggesting and discussing possible approaches. Only provide a detailed solution when you can.

A important objective of working with these problems is to make your thinking visible and to practice your own judgement in this area. *You should therefore not discuss the problems of this module with other groups.*

However, you are encouraged to take the opportunity to search and read about the different models and methods listed above, which are all approaches that you may want to consider when solving new problems. See also the general course information for an explanation of what we expect for your submission.

Predict the temperature

- For the next weekend
- For the same date as today but next year

Bingo lottery problem

In a bingo lottery 1000 lottery tickets are printed. Every ticket consists of a 5x5-square where each of the numbers 1 to 25 are randomly placed. A winning combination is received for a horizontal, vertical or diagonal row. For the draw of the numbers it is desired to select a set of numbers so that exactly 120 tickets give a win (or at least as close as possible), and no ticket must give more than one win. Knowledge of all the lottery tickets can be assumed.

Public transport departure forecast

At the stops for public transport there is often a screen predicting the next departure time for each line. Suggest different approaches for how this time can be estimated.

Film festival problem

Quite a few years ago we had a system at the CSE department creating automatic schedules for film festival enthusiasts, enabling everyone to quickly order tickets before they run out. When the festival programme arrives, all films are quickly entered. Individuals can then indicate desired films with a priority and the system automatically creates an entire schedule for the week. The challenge is that it is usually impossible to see all films since many are shown at the same time, and each film is often shown several times during the week.

Product rating in consumer test

E.g. how determine the score for different dishwashers in a consumer test?

Constraint satisfaction and constraint programming (read and explain)

a) Read about **constraint satisfaction problems** and **constraint programming**. Use any links or material that you find useful and give them as references. Briefly explain the main concepts. For this reading task feel free to discuss with others to help you understand the topics!

(Of course this is not the only important technique, but we assume you already know some things about algorithms, optimization and data science!)

b) (voluntary) Is it possible to generalize *constraint programming (and constraint propagation)* to numbers, and not just true and false? Think and describe your thoughts. *(Explanation: In constraint programming we use a table - or more generally a function - to describe a constraint. For example, if we have two three-valued variables, the constraint can be described by a table with 3*3=9 values, all of which is either T or F. But what would happen if we replaced T and F by numbers of some kind? Another completely different issue is if you can have real numbers as variables in constraint programming and how to handle that - this we not what I intended to ask about but if this is how you interpreted the question, that's fine - this is also an interesting issue.)*

Self-check

- Have you answered all questions to the best of your ability?
- Is all the required information on the front page, is the file name correct etc.?
- Anything else you can easily check? (details, terminology, arguments, clearly stated answers etc.?)

Do not submit an incomplete module! We are available to help you, and you can receive a short extension if you contact us.

	File name	Size	
	Module1Group19.pdf	176 KB	Download

ZOOM

New Attempt